

# MAT3032 ANSWERS TO EXERCISES 3.3

## Exercises 3.3

1. (a)  $x1 = a + bi$ ,  $xi = -b + ai$ , so the matrix representing  $x$  is  $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ .

(b)  $x1 = a + bv$ ,  $xv = av + bv^2 = av$ , so the matrix representing  $x$  is  $\begin{pmatrix} a & 0 \\ b & a \end{pmatrix}$ .

(c)  $xE_{11} = \begin{pmatrix} a & 0 \\ c & 0 \end{pmatrix} = aE_{11} + cE_{21}$ ,  $xE_{12} = \begin{pmatrix} 0 & a \\ 0 & c \end{pmatrix} = aE_{12} + cE_{22}$ ,

$xE_{21} = \begin{pmatrix} b & 0 \\ d & 0 \end{pmatrix} = bE_{11} + dE_{21}$ ,  $xE_{22} = \begin{pmatrix} 0 & b \\ 0 & d \end{pmatrix} = bE_{12} + dE_{22}$ ,

so the matrix representing  $x$  is  $\begin{pmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ c & 0 & d & 0 \\ 0 & c & 0 & d \end{pmatrix}$ .

(d)  $xk = ak - bj + ci - d1$ , etc (as in Example 3.3.2), so the matrix representing

$x$  is  $\begin{pmatrix} a & b & -c & d \\ -b & a & d & c \\ c & -d & a & b \\ -d & -c & -b & a \end{pmatrix}$ .

2. (a)  $\mathbb{F}_3C_2$  has  $3^2 = 9$  elements. Its dimension over  $\mathbb{F}_3$  is 2.

(b)  $\mathbb{F}_2S_3$  has  $2^6 = 64$  elements. Its dimension over  $\mathbb{F}_2$  is 6.

(c)  $\mathbb{F}_5A_4$  has  $5^{12}$  elements. Its dimension over  $\mathbb{F}_5$  is 12.

(d)  $\mathbb{Q}D_4$  is infinite (since  $\mathbb{Q}$  is). Its dimension over  $\mathbb{Q}$  is 8.

3.  $\mathbb{F}_2C_3 = \{a + bg + cg^2 : a, b, c \in \{0, 1\}\}$

$= \{0, 1, g, g^2, 1 + g, 1 + g^2, g + g^2, 1 + g + g^2\}$ .

| +             | 0             | 1             | $g$           | $g^2$         | $1 + g$       | $1 + g^2$     | $g + g^2$     | $1 + g + g^2$ |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| 0             | 0             | 1             | $g$           | $g^2$         | $1 + g$       | $1 + g^2$     | $g + g^2$     | $1 + g + g^2$ |
| 1             | 1             | 0             | $1 + g$       | $1 + g^2$     | $g$           | $g^2$         | $1 + g + g^2$ | $g + g^2$     |
| $g$           | $g$           | $1 + g$       | 0             | $g + g^2$     | 1             | $1 + g + g^2$ | $g^2$         | $1 + g^2$     |
| $g^2$         | $g^2$         | $1 + g^2$     | $g + g^2$     | 0             | $1 + g + g^2$ | 1             | $g$           | $1 + g^2$     |
| $1 + g$       | $1 + g$       | $g$           | 1             | $1 + g + g^2$ | 0             | $g + g^2$     | $1 + g^2$     | $g^2$         |
| $1 + g^2$     | $1 + g^2$     | $g^2$         | $1 + g + g^2$ | 1             | $g + g^2$     | 0             | $1 + g$       | 1             |
| $g + g^2$     | $g + g^2$     | $1 + g + g^2$ | $g^2$         | $g$           | $1 + g^2$     | $1 + g$       | 0             | 1             |
| $1 + g + g^2$ | $1 + g + g^2$ | $g + g^2$     | $1 + g^2$     | $1 + g$       | $g^2$         | $g$           | 1             | 0             |

As it is abelian of order 8 and all elements are self-inverse, we see that  $(\mathbb{F}_2C_3, +) \cong C_2 \times C_2 \times C_2$ . Indeed, writing  $a + bg + cg^2$  as  $(a, b, c)$  gives the table for  $C_2 \times C_2 \times C_2$  as in Coursework 1.

| $\times$  | 1         | $g$       | $g^2$     | $1+g$   | $1+g^2$ | $g+g^2$ | $1+g+g^2$ |
|-----------|-----------|-----------|-----------|---------|---------|---------|-----------|
| 1         | 1         | $g$       | $g^2$     | $1+g$   | $1+g^2$ | $g+g^2$ | $1+g+g^2$ |
| $g$       | $g$       | $g^2$     | 1         | $g+g^2$ | $1+g$   | $1+g^2$ | $1+g+g^2$ |
| $g^2$     | $g^2$     | 1         | $g$       | $1+g^2$ | $g+g^2$ | $1+g$   | $1+g+g^2$ |
| $1+g$     | $1+g$     | $g+g^2$   | $1+g^2$   | $1+g^2$ | $g+g^2$ | $1+g$   | 0         |
| $1+g^2$   | $1+g^2$   | $1+g$     | $g+g^2$   | $g+g^2$ | $1+g$   | $1+g^2$ | 0         |
| $g+g^2$   | $g+g^2$   | $1+g^2$   | $1+g$     | $1+g$   | $1+g^2$ | 0       | 0         |
| $1+g+g^2$ | $1+g+g^2$ | $1+g+g^2$ | $1+g+g^2$ | 0       | 0       | 0       | $1+g+g^2$ |

$(\mathbb{F}_2 C_3^*, \times)$  is not a group.

4. By Proposition 3.9,  $e = \frac{1}{5}(1 + g + g^2 + g^3 + g^4)$  is a central idempotent.

Then  $\mathbb{R}C_5 = e\mathbb{R}C_5 \oplus (1 - e)\mathbb{R}C_5$ . The two ideals have dimensions 1 and 4.

The matrix representing  $e$  relative to  $(1, g, g^2, g^3, g^4)$  is  $\frac{1}{5} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$ .

5.  $f^2 = \frac{1}{9}(1 + \omega g + \omega^2 g^2 + \omega g + \omega^2 g^2 + \omega^3 g^3 + \omega^2 g^2 + \omega^3 g^3 + \omega^4 g^4)$ .

Now  $g^3 = 1$  and  $\omega^3 = 1$  (as  $\omega$  is a complex cube root of unity),

so  $f^2 = \frac{1}{9}(3 + 3\omega g + 3\omega^2 g^2) = f$ . Thus  $f$  is idempotent.

$ef = \frac{1}{9}(1 + g + g^2)(1 + \omega g + \omega^2 g^2) = \frac{1}{9}(1 + \omega g + \omega^2 g^2 + g + \omega g^2 + \omega^2 + g^2 + \omega + \omega^2 g) = \frac{1}{9}(1 + \omega + \omega^2)(1 + g + g^2) = 0$ , since we know that  $1 + \omega + \omega^2 = 0$  in  $\mathbb{C}$ .

Hence  $e$  and  $f$  are orthogonal idempotents in  $A$ .

$ef = 0$ , so  $f = f - ef = (1 - e)f$ . Thus  $f \in (1 - e)A$ .

6. We have  $g^n = 1$ , so  $g^n - 1 = 0$ , so  $(g - 1)(g^{n-1} + \cdots + 1) = 0$ , so  $g - 1$  is a zero-divisor.

A zero-divisor cannot have an inverse, since if  $ab = 0$  and  $a^{-1}$  exists then  $b = 0$ .

Hence  $g - 1$ , which is non-zero, is not a unit, so  $KC_4$  is not a division ring.

7. Suppose  $e$  is a sum of two orthogonal idempotents  $f$  and  $e - f$ .

As  $f \in KG$ , we have  $f = a_1 x_1 + \cdots + a_n x_n$  for some  $a_1, \dots, a_n \in K$ , so  $fe = a_1 x_1 e + \cdots + a_n x_n e = (a_1 + \cdots + a_n)e = ke$  for some  $k \in K$ .

Also  $0 = f(e - f) = fe - f^2 = fe - f$ , so  $fe = f$ . Thus  $f = ke$ .

As  $f$  is idempotent,  $ke = k^2 e^2 = k^2 e$ . If  $k \neq 0$  then  $k^{-1}ke = k^{-1}k^2 e$ , so  $e = ke$ . Thus  $f = 0$  or  $f = e$ .

Hence there are no two orthogonal idempotents, both non-zero, whose sum is  $e$ , so  $e$  is a primitive idempotent.

8. (a)  $|A| = 5^3 = 125$ .

(b)  $(a + bx + cx^2)x = ax + bx^2 + cx = (a + c)x + bx^2$ ,  $(a + bx + cx^2)x^2 = (a + c)x^2 + bx$ , so the matrix is  $\begin{pmatrix} a & 0 & 0 \\ b & a + c & b \\ c & b & a + c \end{pmatrix}$ .

(c) Determinant  $= a((a + c)^2 - b^2) = a(a + c - b)(a + c + b)$ , so  $a + bx + cx^2$  is a unit of  $A$  iff  $a \neq 0$ ,  $a \neq b - c$ ,  $a \neq -b - c$ .

Thus  $a + bx + cx^2$  is a unit if  $a \in \{1, 2, 3, 4\}$ ,  $b \in \{0, 1, 2, 3, 4\}$ ,  $c \notin \{b - a, -b - a\}$ . For each  $a$  we get 4 values of  $c$  with  $b = 0$  and 3 values of  $c$  for each  $b \neq 0$ , so the number of units is  $4(4 + 12) = 64$ .

Now suppose an element is self-inverse, so  $(a + bx + cx^2)^2 = 1$ .

Then  $a^2 + 2b(a + c)x + (b^2 + c^2 + 2ac)x^2 = 1$  so  $a \in \{1, 4\}$  and either  $b = 0$ , so  $c(c + 2a) = 0$ , or  $c = -a$ , so  $b^2 = a^2$ .

Thus the self-inverse elements are

$(1, 0, 0)$ ,  $(1, 0, 3)$ ,  $(1, 1, 4)$ ,  $(1, 4, 4)$ ,  $(4, 0, 0)$ ,  $(4, 0, 2)$ ,  $(4, 1, 1)$ ,  $(4, 4, 1)$ ,

which form a group isomorphic to  $C_2 \times C_2 \times C_2$ .